

## MARSHALLTOWN, IA, USA

WMO: 725461

Lat: 42.111N

Lon: 92.916W

Elev: 974

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94988

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.3       | -2.9 | -14.4                         | 2.6 | -7.8 | -9.3 | 3.5 | -2.0 | 33.4                  | 22.4 | 29.1 | 17.8 | 8.4                   | 310  | 0.593 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.4                   | 90.3            | 76.5 | 87.9 | 75.0 | 84.9 | 72.9 | 78.7                | 87.1 | 77.1 | 85.1 | 75.4 | 82.5 | 11.8                 | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.3                            | 142.4 | 84.2 | 74.6 | 134.4 | 82.4 | 72.8 | 126.2 | 80.4 | 43.0          | 87.5 | 41.2 | 85.1 | 39.5 | 83.2 | 84.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |     |     |
| (4) 27.8          | 24.7 | 21.5 | -16.3                      | 94.1 | 5.4                | 2.9 | -20.2                                              | 96.2 | -23.4      | 97.9 | -26.4      | 99.5 | -30.4      | 101.6 |     |     |
| (5)               |      |      | -16.6                      | 82.1 | 5.3                | 1.6 | -20.5                                              | 83.3 | -23.6      | 84.2 | -26.6      | 85.2 | -30.5      | 86.4  | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |                            | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|----------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |                            | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg                      | 48.5        | 20.1  | 23.5  | 36.0  | 49.1  | 60.6  | 70.0  | 73.3  | 70.5  | 64.1  | 50.9  | 37.5  | 25.0  |  |  |
|                                                                        | DBStd                      | 21.14       | 12.54 | 12.76 | 12.02 | 9.84  | 8.61  | 6.17  | 5.74  | 5.49  | 8.09  | 9.82  | 10.55 | 11.99 |  |  |
|                                                                        | HDD50                      | 3558        | 928   | 743   | 459   | 131   | 14    | 0     | 0     | 0     | 4     | 109   | 393   | 777   |  |  |
|                                                                        | HDD65                      | 6799        | 1393  | 1162  | 900   | 484   | 190   | 23    | 6     | 13    | 113   | 450   | 824   | 1241  |  |  |
|                                                                        | CDD50                      | 3014        | 0     | 1     | 26    | 103   | 343   | 599   | 723   | 637   | 426   | 136   | 19    | 1     |  |  |
|                                                                        | CDD65                      | 780         | 0     | 0     | 1     | 6     | 54    | 172   | 264   | 185   | 86    | 12    | 0     | 0     |  |  |
|                                                                        | CDH74                      | 7113        | 0     | 0     | 11    | 95    | 506   | 1551  | 2489  | 1534  | 808   | 117   | 2     | 0     |  |  |
|                                                                        | CDH80                      | 2213        | 0     | 0     | 0     | 19    | 130   | 492   | 869   | 436   | 242   | 25    | 0     | 0     |  |  |
|                                                                        | WSAvg                      | 10.3        | 11.6  | 11.6  | 12.0  | 12.7  | 11.6  | 9.4   | 7.5   | 7.0   | 8.4   | 10.0  | 11.1  | 11.1  |  |  |
|                                                                        | PrecAvg                    | 34.1        | 0.9   | 1.1   | 2.2   | 3.2   | 4.1   | 4.6   | 4.0   | 4.7   | 3.7   | 2.5   | 1.7   | 1.3   |  |  |
| (15) Precipitation                                                     | PrecMax                    | 48.0        | 3.8   | 3.5   | 4.5   | 6.4   | 10.0  | 11.3  | 11.3  | 14.9  | 8.9   | 5.7   | 5.6   | 3.4   |  |  |
|                                                                        | PrecMin                    | 17.2        | 0.1   | 0.1   | 0.4   | 0.9   | 0.9   | 1.1   | 0.8   | 0.5   | 0.3   | 0.1   | 0.0   | 0.2   |  |  |
|                                                                        | PrecStd                    | 7.6         | 0.8   | 0.9   | 1.2   | 1.5   | 2.5   | 2.3   | 2.2   | 3.2   | 2.4   | 1.7   | 1.3   | 0.8   |  |  |
|                                                                        |                            |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%                       | DB          | 53.1  | 60.0  | 75.0  | 82.6  | 88.8  | 91.3  | 94.2  | 91.4  | 90.3  | 83.3  | 71.5  | 58.7  |  |  |
|                                                                        |                            | MCWB        | 42.8  | 50.4  | 61.1  | 63.8  | 71.8  | 76.0  | 78.8  | 75.9  | 73.0  | 68.8  | 58.0  | 53.4  |  |  |
|                                                                        | 2%                         | DB          | 44.8  | 52.0  | 66.5  | 76.1  | 83.3  | 88.4  | 90.7  | 88.0  | 86.3  | 76.9  | 63.9  | 50.4  |  |  |
|                                                                        |                            | MCWB        | 38.9  | 46.1  | 55.8  | 60.6  | 67.6  | 74.0  | 78.6  | 76.3  | 71.9  | 63.5  | 55.1  | 45.9  |  |  |
|                                                                        | 5%                         | DB          | 39.2  | 45.4  | 60.8  | 71.4  | 79.7  | 85.6  | 88.0  | 84.4  | 82.2  | 72.3  | 58.4  | 45.5  |  |  |
|                                                                        |                            | MCWB        | 35.7  | 40.9  | 53.3  | 57.7  | 64.9  | 71.5  | 76.9  | 74.3  | 69.4  | 62.0  | 50.9  | 41.2  |  |  |
|                                                                        | 10%                        | DB          | 35.8  | 39.3  | 54.2  | 66.1  | 75.4  | 82.3  | 84.6  | 81.8  | 78.9  | 67.2  | 53.9  | 39.9  |  |  |
|                                                                        |                            | MCWB        | 33.4  | 36.1  | 47.1  | 54.6  | 63.1  | 70.1  | 74.6  | 72.4  | 67.2  | 58.8  | 47.5  | 36.7  |  |  |
|                                                                        | 0.4%                       | WB          | 44.3  | 52.9  | 62.9  | 66.2  | 73.5  | 79.5  | 82.2  | 79.7  | 76.7  | 71.2  | 60.4  | 54.8  |  |  |
|                                                                        |                            | MCDB        | 50.6  | 58.8  | 71.9  | 78.0  | 85.4  | 89.0  | 90.1  | 87.8  | 85.8  | 78.8  | 66.9  | 58.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%                         | WB          | 39.6  | 46.3  | 58.8  | 62.5  | 70.8  | 76.6  | 79.8  | 77.5  | 74.4  | 67.4  | 56.5  | 47.6  |  |  |
|                                                                        |                            | MCDB        | 44.2  | 50.6  | 64.5  | 72.6  | 80.8  | 85.0  | 88.1  | 85.5  | 82.4  | 72.9  | 62.9  | 49.1  |  |  |
|                                                                        | 5%                         | WB          | 36.1  | 40.9  | 53.1  | 59.4  | 68.0  | 74.6  | 78.0  | 75.7  | 72.1  | 63.7  | 52.2  | 41.3  |  |  |
|                                                                        |                            | MCDB        | 39.1  | 45.3  | 59.8  | 68.9  | 76.0  | 82.6  | 86.1  | 82.5  | 79.0  | 69.2  | 57.7  | 45.0  |  |  |
|                                                                        | 10%                        | WB          | 33.4  | 36.4  | 47.5  | 55.9  | 65.6  | 72.3  | 76.3  | 74.0  | 69.6  | 59.9  | 47.6  | 37.1  |  |  |
|                                                                        |                            | MCDB        | 35.3  | 39.4  | 53.5  | 64.2  | 73.0  | 80.2  | 83.6  | 80.3  | 75.5  | 66.6  | 52.9  | 40.1  |  |  |
|                                                                        | 5% DB                      | MDBR        | 16.0  | 16.8  | 18.5  | 21.6  | 20.5  | 19.6  | 19.4  | 19.6  | 22.7  | 21.0  | 18.6  | 15.8  |  |  |
|                                                                        |                            | MCDBR       | 20.4  | 22.1  | 27.5  | 29.4  | 24.6  | 22.6  | 20.7  | 21.6  | 25.2  | 26.3  | 25.8  | 21.0  |  |  |
|                                                                        |                            | MCWBR       | 15.9  | 16.1  | 17.2  | 16.4  | 12.3  | 11.3  | 11.0  | 11.6  | 12.3  | 14.5  | 17.1  | 16.1  |  |  |
|                                                                        |                            | MCWBR       | 19.1  | 21.5  | 24.8  | 24.5  | 20.9  | 19.8  | 19.0  | 19.2  | 20.9  | 20.9  | 23.5  | 19.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB                      | MCWBR       | 15.2  | 16.2  | 16.2  | 15.7  | 11.8  | 11.8  | 11.1  | 11.1  | 11.7  | 14.0  | 17.0  | 15.7  |  |  |
|                                                                        |                            |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        | Clear-Sky Solar Irradiance | taub        | 0.284 | 0.304 | 0.329 | 0.377 | 0.407 | 0.420 | 0.435 | 0.415 | 0.385 | 0.340 | 0.301 | 0.281 |  |  |
|                                                                        |                            | taud        | 2.401 | 2.349 | 2.386 | 2.279 | 2.264 | 2.277 | 2.214 | 2.284 | 2.322 | 2.445 | 2.483 | 2.433 |  |  |
| (42) All-Sky Solar Radiation                                           | RadAvg                     | Ebn at Noon | 275   | 286   | 291   | 283   | 276   | 272   | 266   | 268   | 268   | 269   | 267   | 266   |  |  |
|                                                                        |                            | Edh at Noon | 25    | 31    | 34    | 41    | 42    | 42    | 44    | 40    | 36    | 28    | 23    | 22    |  |  |
| (44) All-Sky Solar Radiation                                           | RadStd                     |             | 597   | 921   | 1210  | 1495  | 1768  | 1958  | 1988  | 1721  | 1432  | 948   | 642   | 484   |  |  |
|                                                                        |                            |             | 59    | 69    | 120   | 125   | 128   | 178   | 128   | 110   | 103   | 116   | 53    | 40    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (45 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page